

Beyond Vector Search

GraphRAG vs VectorRAG vs HybridRAG: A Multi-Model
Benchmarking Study with Fine-Tuning and
Prompt Engineering Analysis

Project Report

Project Type	RAG Benchmarking and Analytics System
Main Components	GraphRAG, VectorRAG, HybridRAG, Evaluation, Streamlit Dashboard, Docker
Dataset Size	70 benchmark questions
Benchmark Runs	1,120 evaluated runs
Dashboard	Streamlit-based benchmark analytics
Deployment	Docker and Docker Compose

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1. Introduction

Retrieval-Augmented Generation has become one of the most practical approaches for building question-answering systems over private or domain-specific data. Instead of relying only on the internal knowledge of a language model, a RAG system retrieves external evidence and uses that evidence to generate a grounded answer.

However, not all RAG systems retrieve information in the same way. A vector-based RAG system is strong at semantic similarity, but it may struggle with relationship-heavy questions. A graph-based RAG system can retrieve structured relationships, but it depends heavily on schema quality and query generation. A hybrid system can combine both approaches.

This project was built to compare these retrieval strategies in a measurable way. The system benchmarks VectorRAG, GraphRAG, HybridRAG, and an optimized GraphRAG V2 pipeline across multiple answer models. The project also includes structured evaluation, hallucination scoring, latency tracking, retrieval evidence inspection, generated Cypher analysis, a Streamlit dashboard, and Dockerized dashboard deployment.

The goal was not only to build a working RAG application, but to evaluate which pipeline performs better, why it performs better, and where each approach fails.

2. System Architecture

The system is divided into five main layers:

1. Data layer
2. Retrieval layer
3. Generation layer
4. Evaluation layer
5. Dashboard layer

The data layer stores structured and unstructured information about billionaire profiles, companies, industries, roles, and relationship facts. The graph database stores entities and relationships, while vector retrieval is used for semantic evidence retrieval.

The retrieval layer contains three main pipelines:

- VectorRAG
- GraphRAG
- HybridRAG

The generation layer uses different answer models, including cloud models, local models, and fine-tuned local models.

The evaluation layer scores each generated answer using structured metrics such as correctness, faithfulness, evidence quality, hallucination score, refusal score, and overall score.

The dashboard layer provides an interactive Streamlit interface for analyzing benchmark results, comparing models, reviewing generated Cypher, inspecting retrieval evidence, and checking latency and resource readings.

2.1 High-Level Architecture

The benchmark runner sends each question through one selected retrieval pipeline. The selected pipeline can be VectorRAG, GraphRAG, or HybridRAG.

VectorRAG retrieves semantic evidence from the vector store. GraphRAG generates a Cypher query and retrieves structured evidence from Neo4j. HybridRAG runs both vector retrieval and graph retrieval, then combines both evidence sources before sending the context to the answer model.

HybridRAG is included as a separate retrieval strategy because it uses both semantic retrieval and graph-based retrieval. This allows the system to answer both general semantic questions and relationship-heavy questions more reliably.

2.2 Role of Each Architecture Component

Table 1 explains the role of each major component in the system architecture.

Table 1: Role of major architecture components

Component	Role
Benchmark Runner	Executes the same 70 questions across different pipeline and model configurations.
VectorRAG	Retrieves semantic evidence using vector similarity search. It is useful for direct and concept-based questions.
GraphRAG	Generates Cypher queries and retrieves structured relationship evidence from Neo4j.
HybridRAG	Combines vector retrieval and graph retrieval, then fuses both evidence sources before answer generation.
Answer Model	Generates the final answer using the retrieved context.
Structured Evaluation	Different cloud, local, and fine-tuned models are compared. Scores each answer using correctness, faithfulness, evidence quality, hallucination, refusal, and overall score.
Benchmark CSV	Stores all benchmark outputs, scores, generated Cypher, evidence, latency, and metadata.

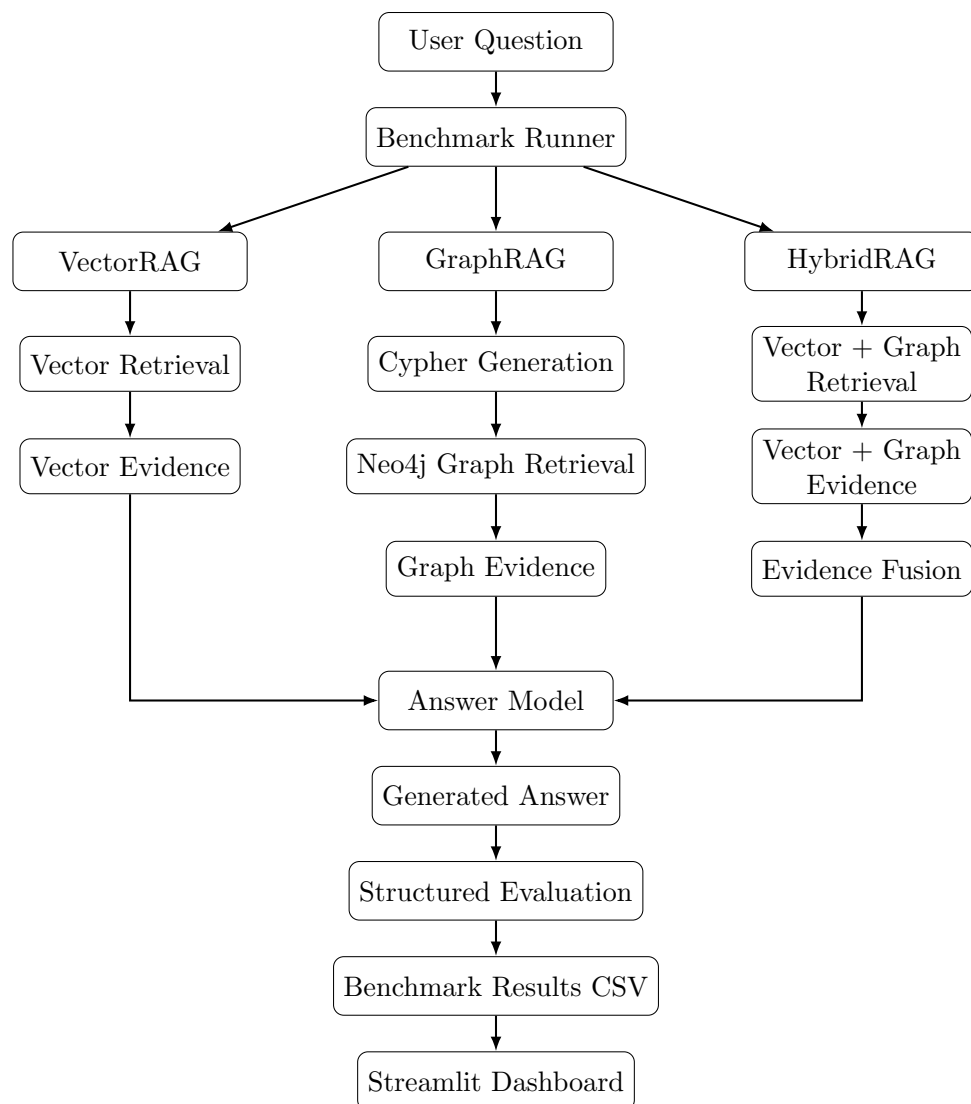


Figure 1: High-level architecture of the Adaptive GraphRAG benchmarking system

3. Dataset

The benchmark uses a fixed evaluation dataset of 70 questions. The questions were designed to test different types of retrieval behavior.

The dataset includes:

- Direct factual questions
- Relationship-based questions
- Multi-hop questions
- Semantic questions
- Numerical questions
- Adversarial or false-claim questions
- Unanswerable questions

Each question includes metadata such as:

- Question ID
- Category
- Question text
- Whether the question is answerable
- Expected evidence type

The purpose of using a fixed dataset was to make the pipeline comparison fair. Every model-pipeline configuration was tested on the same set of questions.

The benchmark results are stored in:

`reports/benchmark_results.csv`

The final cleaned benchmark contains 1,120 evaluated runs.

4. Three Pipelines

4.1 VectorRAG

VectorRAG retrieves context using semantic similarity. The user question is embedded and compared against stored document embeddings. The most relevant chunks are retrieved and passed to the answer model.

VectorRAG performed well for semantic and direct factual questions. It was reliable when the answer could be found through text similarity. However, it was weaker for questions that required relationship traversal or structured multi-hop reasoning.

4.2 GraphRAG

GraphRAG retrieves evidence from a Neo4j graph database. The system generates a Cypher query from the user question, executes the query, and passes the graph evidence to the answer model.

GraphRAG is useful for relationship-heavy questions because graph databases naturally represent connections between people, companies, roles, and industries. However, GraphRAG is also fragile. If the generated Cypher is incorrect, too broad, or not aligned with the schema, the retrieved evidence becomes weak.

4.3 HybridRAG

HybridRAG combines vector evidence and graph evidence. It retrieves semantic context from vector search and structured relationship evidence from the graph database.

HybridRAG was the most stable pipeline overall. It performed well because it did not depend on only one retrieval method. When vector retrieval was not enough, graph evidence helped. When graph retrieval was weak, vector evidence still provided useful context.

4.4 Optimized GraphRAG V2

GraphRAG V2 is an improved version of the original GraphRAG pipeline. The main improvement was the Cypher-generation prompt. The optimized prompt added stronger schema constraints, safer variable rules, relationship-fact-first retrieval, semantic query rules, multi-hop handling, adversarial handling, and few-shot Cypher examples.

OpenAI GraphRAG Before vs After Prompt Optimization						
config	questions	avg_overall_score	avg_correctness	avg_faithfulness	avg_evidence	avg_hallucination
GraphRAG Original + OpenAI	70	2.914	2.943	4.729	3.329	1.271
GraphRAG V2 + OpenAI	70	4.886	4.886	4.957	4.9	1.043

Figure 2: OpenAI GraphRAG original versus GraphRAG V2 comparison

5. Models

The benchmark compared several answer models:

- OpenAI
- NVIDIA
- Groq
- Local Llama 3.2 1B
- Fine-tuned Local Llama 3.2 1B

The reason for testing multiple models was to understand how much answer quality depends on the generator model and how much depends on the retrieval pipeline.

The local model was included to test low-cost and offline inference. The fine-tuned local model was included to compare whether task-specific fine-tuning improves structured answer quality.

6. Evaluation Methodology

Each benchmark run generated one answer for one question using one pipeline and one answer model.

For every run, the system stored:

- Question
- Pipeline name
- Answer model
- Generated answer
- Generated Cypher
- Graph evidence

- Vector evidence
- Evaluation scores
- Error fields
- Batch metadata
- Latency
- Resource readings

The main evaluation metrics were:

Table 2: Evaluation metrics used in the benchmark

Metric	Description
Correctness Score	Measures whether the answer is factually correct.
Faithfulness Score	Measures whether the answer is grounded in retrieved evidence.
Evidence Score	Measures whether the retrieved evidence supports the answer.
Hallucination Score	Measures unsupported or fabricated answer behavior.
Refusal Score	Measures whether unanswerable questions are handled correctly.
Overall Score	Combined quality score.
Local Latency	Time taken to generate the answer locally.

The evaluation focused mainly on answer quality and evidence grounding. CPU and RAM readings were collected for observability, but they were not treated as exact process-level benchmark measurements.

Name	Most Recent Run (7d)	Trace Count (7d)	Error Rate (7d)	P50 Latency (7d)	P99 Latency (7d)	Total Tokens (7d)	Total Cost (7d)
adaptive-graphrag-hybrid-groq	2 days ago	500	~0%	1.96s	6.65s	397,374	\$0.06
adaptive-graphrag-graph-groq	2 days ago	278	~0%	2.40s	7.96s	302,825	\$0.06
adaptive-graphrag-vector-groq	2 days ago	280	0%	1.54s	3.30s	178,256	\$0.03
adaptive-graphrag-hybrid-nvidia	2 days ago	487	~0%	2.40s	42.74s	385,397	\$0.06
adaptive-graphrag-vector-nvidia	2 days ago	313	4%	2.50s	38.67s	176,440	\$0.03
adaptive-graphrag-hybrid-openai	2 days ago	487	~0%	2.09s	7.65s	383,364	\$0.07
adaptive-graphrag-graph-openai	2 days ago	280	0%	2.35s	7.06s	305,601	\$0.06
adaptive-graphrag-vector-openai	2 days ago	280	0%	1.64s	3.58s	175,693	\$0.04

Figure 3: LangSmith trace summary for cloud model RAG runs

Name	Most Recent Run (7d)	Trace Count (7d)	Error Rate (7d)	P50 Latency (7d)	P99 Latency (7d)	Total Tokens (7d)
adaptive-graphrag-graph-nvidia	12 hours ago	276	3%	3.88s	97.03s	301,720
adaptive-graphrag-graph-local-llama3-2-1b-ft	13 hours ago	270	7%	5.23s	16.36s	231,495
adaptive-graphrag-graph-llama3.2:1b	19 hours ago	280	0%	7.06s	51.16s	269,953
adaptive-graphrag-graph-v2-openai	21 hours ago	280	0%	2.35s	6.65s	441,643
adaptive-graphrag-hybrid-local-llama3-2-1b-ft	a day ago	493	0%	2.45s	17.03s	401,825
adaptive-graphrag-vector-local-llama3-2-1b-ft	a day ago	280	0%	2.50s	9.54s	179,035
adaptive-graphrag-hybrid-llama3.2:1b	a day ago	528	1%	3.05s	95.11s	417,649
adaptive-graphrag-vector-llama3.2:1b	a day ago	280	0%	2.35s	18.82s	178,425

Figure 4: LangSmith trace summary for graph and local model runs

7. Results

The final benchmark showed clear differences between retrieval pipelines and model configurations.

Table 3: Average benchmark score by configuration

Configuration	Average Score
HybridRAG + NVIDIA	4.93 / 5
HybridRAG + Groq	4.89 / 5
GraphRAG V2 + OpenAI	4.89 / 5
VectorRAG + NVIDIA	4.79 / 5
VectorRAG + Groq	4.71 / 5
HybridRAG + OpenAI	4.59 / 5
VectorRAG + OpenAI	4.44 / 5
VectorRAG + Fine-tuned Local LLM	4.31 / 5
HybridRAG + Fine-tuned Local LLM	3.64 / 5
HybridRAG + Local LLM	3.40 / 5
Original GraphRAG + OpenAI	2.91 / 5

The strongest overall configuration was HybridRAG with NVIDIA. HybridRAG with Groq and GraphRAG V2 with OpenAI also performed strongly.

The original GraphRAG pipeline performed poorly compared to HybridRAG and VectorRAG. After improving the Cypher generation prompt, GraphRAG V2 improved significantly.

7.1 Dashboard Result Screenshots

config	runs	questions	avg_overall_score	avg_correctness	avg_faithfulness	avg_evidence	avg_hallucination	avg_latency_seconds
HybridRAG + NVIDIA	70	70	4.929	4.929	4.929	4.929	0.986	25.483
HybridRAG + Groq	70	70	4.886	4.886	4.971	4.914	1.029	7.619
GraphRAG V2 + OpenAI	70	70	4.886	4.886	4.957	4.9	1.043	4.338
VectorRAG + NVIDIA	70	70	4.786	4.771	4.971	4.857	1.029	14.836
VectorRAG + Groq	70	70	4.714	4.714	5	4.829	1	2.823
HybridRAG + OpenAI	70	70	4.586	4.586	4.929	4.586	0.986	9.125
VectorRAG + OpenAI	70	70	4.443	4.443	4.986	4.457	1.014	3.295
VectorRAG + Local Llama 3.2 1B FT	70	70	4.314	4.286	4.529	4.4	1.471	9.745
HybridRAG + Local Llama 3.2 1B FT	70	70	3.643	3.657	4.343	4.057	1.657	19.738
HybridRAG + Local Llama 3.2 1B	70	70	3.4	3.414	4.629	4.257	1.286	35.148

Figure 5: Benchmark summary table for top-performing configurations

config	runs	questions	avg_overall_score	avg_correctness	avg_faithfulness	avg_evidence	avg_hallucination	avg_latency_seconds
VectorRAG + OpenAI	70	70	4.443	4.443	4.986	4.457	1.014	3.295
VectorRAG + Local Llama 3.2 1B FT	70	70	4.314	4.286	4.529	4.4	1.471	9.745
HybridRAG + Local Llama 3.2 1B FT	70	70	3.643	3.657	4.343	4.057	1.657	19.738
HybridRAG + Local Llama 3.2 1B	70	70	3.4	3.414	4.629	4.257	1.286	35.148
GraphRAG Original + OpenAI	70	70	2.914	2.943	4.729	3.329	1.271	4.236
GraphRAG + Groq	70	70	2.757	2.743	4.586	3.071	1.329	4.908
GraphRAG + NVIDIA	70	70	2.586	2.586	4.314	2.8	1.343	20.334
GraphRAG + Local Llama 3.2 1B	70	70	2.386	2.529	3.586	2.443	2.414	21.414
GraphRAG + Local Llama 3.2 1B FT	70	70	2.314	2.486	2.414	2.4	2.729	10.088
VectorRAG + Local Llama 3.2 1B	70	70	2.271	2.271	4.829	3.643	1.171	15.082

Figure 6: Benchmark summary table for remaining configurations

8. Key Findings

8.1 HybridRAG was the most stable pipeline

HybridRAG performed strongly because it combined semantic retrieval and graph retrieval. This made it more robust across different question types.

8.2 GraphRAG depends heavily on Cypher quality

The first GraphRAG implementation underperformed because generated Cypher queries were often incomplete, too broad, or not aligned with the graph schema.

8.3 GraphRAG V2 showed major improvement

After optimizing the Cypher generation prompt, OpenAI GraphRAG improved from around 2.91 / 5 to around 4.89 / 5.

8.4 VectorRAG was reliable but limited

VectorRAG worked well for semantic questions, but it was weaker when the question required explicit relationship reasoning.

8.5 Evaluation requires more than correctness

Correctness alone did not explain pipeline behavior. Faithfulness, hallucination score, evidence usage, generated Cypher, and retrieval evidence were necessary to understand why a system succeeded or failed.

9. Prompt Engineering Experiment

The most important experiment in the project was the GraphRAG prompt optimization experiment.

The original Cypher generation prompt was functional but too general. It gave schema rules and basic routing instructions, but it did not strongly control semantic queries, multi-hop questions, adversarial questions, or relationship-fact-first retrieval.

The optimized GraphRAG V2 prompt added:

- Stricter schema constraints
- Fixed return format
- Safer variable usage
- RelationshipFact-first retrieval
- Semantic query rules
- Multi-hop query handling
- Competition query handling
- Numerical investment query handling
- Adversarial question handling
- Few-shot Cypher examples
- Alias rules for entity matching

This improved GraphRAG performance significantly.

Table 4: GraphRAG prompt optimization result

Version	Average Score
Original GraphRAG + OpenAI	2.91 / 5
GraphRAG V2 + OpenAI	4.89 / 5

The full before/after prompt comparison is documented separately in:

`src/graphs/prompts.py`

This experiment showed that GraphRAG quality is not only about having a graph database. It also depends on how accurately the system converts natural language into schema-valid Cypher queries.

10. Fine-Tuning Analysis

The project also compared a base local LLM with a fine-tuned local LLM.

The base local model was useful for testing offline and low-cost inference, but it performed weaker than cloud models. The fine-tuned local model improved performance in some configurations, especially VectorRAG.

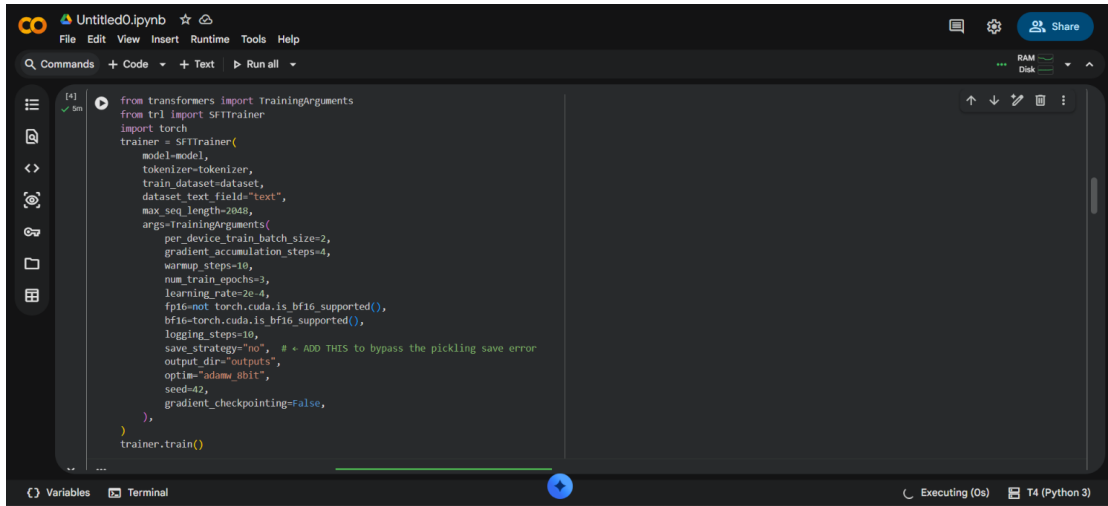


Figure 7: Loading the local model with Unsloth in Google Colab

The fine-tuned local model result showed that smaller models can become more useful when adapted to the task format. However, fine-tuning did not automatically make the local model better than larger cloud models.

Main observations:

- Fine-tuning improved local model behavior in VectorRAG.
- Local fine-tuned performance was still limited by model size.
- HybridRAG with the fine-tuned model did not outperform the best cloud-model configurations.
- Fine-tuning was useful, but retrieval quality still had a major effect on answer quality.

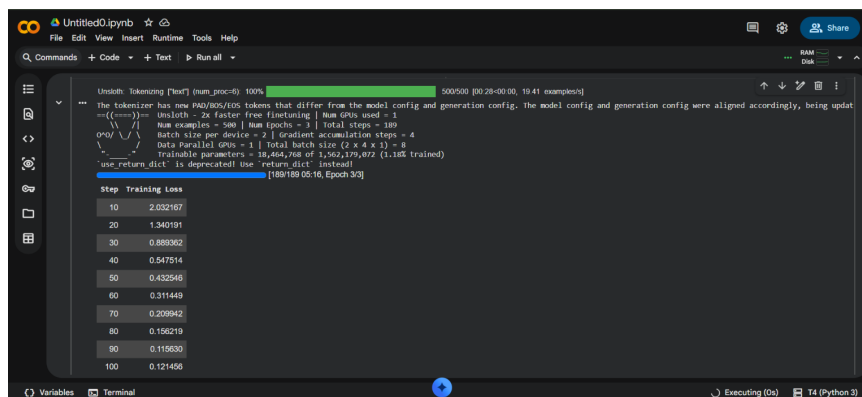


Figure 8: Fine-tuning local LLM using Unsloth in Google Colab

This suggests that fine-tuning and retrieval design should be treated as complementary. A better generator cannot fully compensate for poor retrieval evidence.

11. Limitations

This project has several limitations:

1. The benchmark uses a fixed set of 70 questions.
2. Results are specific to the dataset, graph schema, and evidence structure used in this project.
3. GraphRAG performance depends heavily on Cypher generation quality.
4. CPU and RAM readings are observability snapshots, not exact process-level resource measurements.
5. Token and cost analysis can be expanded further with complete provider-level or LangSmith exports.
6. Local model results depend on hardware, inference setup, and model size.
7. The fine-tuning analysis is limited to the selected local model and task setup.

12. Future Work

Future improvements can include:

- Add complete token and cost tracking.
- Add process-level CPU and peak RAM monitoring.
- Add more benchmark questions.
- Add automatic error category analysis.
- Add retrieval metrics such as Recall@K, MRR, and hit rate.
- Add CI-based benchmark regression testing.
- Add more local models.
- Add a larger fine-tuned model.
- Add live RAG testing inside the dashboard.
- Add Neo4j and dashboard together in Docker Compose.
- Add support for real-time query routing between VectorRAG, GraphRAG, and HybridRAG.

13. Conclusion

This project set out to answer a practical question: does the choice of retrieval strategy actually matter in a RAG system, or is the language model doing most of the work?

The benchmark results across 1,120 evaluations gave a clear answer. Retrieval strategy matters significantly. HybridRAG consistently outperformed both VectorRAG and GraphRAG across all five models tested, confirming that combining semantic search with structured graph retrieval produces more complete and reliable answers than either approach alone.

GraphRAG was the most revealing finding. Initial results showed it scoring 2.91 on average — the weakest pipeline across all models. Rather than accepting this as a fundamental limitation of graph-based retrieval, the Cypher generation prompt was redesigned with explicit relationship traversal patterns, fuzzy entity matching, and few-shot examples. The score jumped to 4.89 on the same model and same dataset. No architecture change. No retraining. Just a better prompt. This single experiment demonstrated that evaluation without iteration is incomplete — a low score tells you something is wrong, not that the approach is wrong.

The fine-tuning experiment produced an equally important finding. A Llama 1B model running locally on a CPU scored 2.27 on VectorRAG before fine-tuning. After domain-specific fine-tuning using Unsloth and QLoRA on a free Colab GPU, the same model scored 4.31 — within 0.13 points of GPT-4o Mini at zero inference cost. This suggests that for domain-specific applications, fine-tuning a small local model is a more cost-efficient path than defaulting to large commercial APIs.

Three lessons stand out from this work. First, GraphRAG requires careful schema design and prompt engineering to be useful — it is not a drop-in improvement over vector search. Second, evaluation frameworks must go beyond accuracy scores. Retrieval precision, and resource monitoring revealed failure modes that overall scores would have hidden. Third, model size matters less than expected when fine-tuning is applied correctly.

A strong RAG system is not built by picking the right model. It is built by understanding where each retrieval strategy fails, measuring those failures precisely, and iterating on the components that cause them.

14. References

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